Federica Padovano

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Languages: Italian, English, French

Education

10/2023-09/2026 — **LJLL, Sorbonne Université (SU)**
PhD in Applied Mathematics.
Title: "Mathematical models of tumour evolutionary dynamics and tumour-immune interactions: understanding the role of heterogeneity in therapeutic outcomes"
Advisors: Prof. L. Almeida (LPSM, SU), Prof. M. Cerezo (Université Côte d'Azur), Dr. C. Villa (MAP5, UPC)

09/2021-07/2023 — **École Polytechnique Fédérale de Lausanne (EPFL)**
MSc in Applied Mathematicas.
Master project: "An evolutionary dynamics model of the development of drug resistance in metastatic tumors" (LJLL, Sorbonne Université, Paris)
Advisors: Prof. L. Almeida, Prof. M. Picasso (EPFL), Dr. C. Villa

09/2018-07/2021 — **Politecnico di Torino (PoliTo)**
BSc in Mathematics For Engineering.
Bachelor project: "Study of an aggregation and propagation model of amyloid in neurons to describe the progression of Alzheimer's disease"
Advisor: Prof. A. Tosin (PoliTo)

Industry experience

09/2022-02/2023 — **Novadiscovery**, Lyon
Biomodeler internship, "Physiologically-based pharmacokinetic model for small molecule drugs"

Publications & Preprints

F. Padovano, C. Villa, *The development of drug resistance in metastatic tumours under chemotherapy: an evolutionary perspective*, Journal of Theoretical Biology, 595(1):111957, 2024, https://doi.org/10.1016/j.jtbi.2024.111957.

R. Singh, J. Glowacki, M. Beaud, F. Padovano, R.S. Manning, J.H. Maddocks, *cgNA+min: computation of sequence-dependent dsDNA energy-minimizing minicircles*, Nucleic Acids Research, 54(9), 2026, https://doi.org/10.1093/nar/gkag398

F. Padovano, , A. M. Lombardi (UniTo), S. Novello (UniTo), T. Lorenzi (PoliTo), E. Vigna (UniTo), E. Leschiera (ESILV), C. Villa, *Modelling the efficacy of CIK cell immunotherapy targeting MET-expressing mesothelioma cells*, Scientific Reports, 2026, https://doi.org/10.1038/s41598-026-64219-y

Ongoing projects

F. Padovano, L. Almeida, A. C. Bastos (Université Côte d'Azur), M. Cerezo, C. Villa, *Mathematical modelling of the role of nutrients in the tumour microenvironment on the efficacy of melanoma treatment*

Meetings with contribution

07/2026, ECMTB	**Deterministic models to describe and analyse tumour-immune interactions and immunotherapy** (Conference - Minisymposium speaker)
06/2026, Paris (LJLL)	**Workshop on PDEs and Mathematical Biology** (Workshop - Invited Speaker)
06/2026, Nanterre	**Collective Behavior in Complex Systems** (Conference - Keynote Invited Speaker)
05/2026, Oxford	**Mathematical Oncology Group meeting** (Visiting period - Invited speaker)
02/2026, PoliTo	**From Micro To Macro in Mathematical Biology: Recent advances in the modeling and analysis of biological systems** (Workshop - Contributed talk)
11/2025, Grenoble	**J-Biot 2025: Les 1ères Journées de la Biologie Théorique** (Workshop - Contributed talk)
09/2025, SIMAI	**Mathematical and Computational Models for Tumor Evolution and Therapy** (Conference minisymposium - Invited speaker)
06/2025, SMAI	**Mathematical advances in modelling cancer treatment** (Conference minisymposium - Invited speaker)
04/2025, Bonn	**Young women in mathematical biology** (Workshop - Invited speaker)

Event organisation

14-16/10/2026	Workshop "**Data-driven approaches in modelling cell and tissue dynamics**", LJLL, Paris (https://datadrivenmodel.sciencesconf.org/)
07/2026	Minisymposium "**Deterministic models to describe and analyse tumour-immune interactions and immunotherapy**", ECMTB, Graz

Grants

04/2025	European Women in Mathematics (EWM) travel grant (400€)
10/2025	BOUM SMAI funding for workshop organisation in 09/2026 (500€)
04/2026	European Society of Mathematical and Theoretical Biology (ESMTB) travel grant (350€)
06/2026	BioConvergence pour Santé (BioConvS) funding for workshop organisation (1000€)

Teaching experience

2023-2026 **Teaching assistant, Sorbonne Université**

- 54h (1 semester) of the BSc course *Mathematics for sciences* 1 (Basic notions: calculus, differential equations, multivariable functions, and linear transformations of the plane)
- 138h (3 semesters) of the BSc course *Mathematics for sciences* 2 (Linear algebra and Probability)